

AI CAREER ROADMAP ANALYZER

Dashboard Insights, Analysis & Strategic Recommendations Report

Prepared as an independent data-analyst review of the Power BI project

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1. Executive Summary

The AI Career Roadmap Analyzer is a six-page Power BI dashboard that combines global AI job-market analytics with a personalized career-guidance layer. It covers hiring trends, salary benchmarks, in-demand roles, a structured skills roadmap, forward-looking industry trends, and tailored career recommendations, drawing on a dataset spanning work years 2020–2025 across 97 companies and roughly 151,000 job records.

The project succeeds in its core goal: it turns a raw jobs dataset into an accessible, guided narrative for someone planning a career in AI. The visual identity is consistent, the navigation is intuitive, and the roadmap page in particular is a genuinely useful, differentiated feature that most "job market dashboard" projects skip.

At the same time, a close read of the KPI cards and chart aggregations across pages surfaces several data-consistency issues that should be fixed before this is presented as a portfolio or interview piece — most notably mismatched "Total Jobs" figures between pages, a salary KPI that appears to be summing rather than averaging, and an unrealistic salary outlier (30M) that is dragging the "Highest Salary" card. Section 6 documents these in detail, and Section 7 gives a prioritized fix list.

- **Bottom line:** this is a strong, well-designed final-year project. Tightening the data model and aggregation logic will take it from "visually polished" to "analytically rigorous" — the difference that stands out in interviews and evaluations.

2. Project Overview

2.1 Purpose

To help aspiring and current professionals explore the global AI job market — in-demand roles, required skills, salary trends, hiring patterns — and to translate that market data into a structured, personalized learning path ("Explore • Learn • Grow • Succeed").

2.2 Structure (6 pages)

- Overview — project introduction, headline KPIs, and feature summary
- AI Job Market Overview — global job distribution, employee-type mix, salary trend by year
- AI Career Roadmap — Beginner / Intermediate / Advanced skill path, most in-demand roles, hiring trend
- Salary Analysis — salary by experience level, by job role, by geography, and salary growth over time
- AI Future Trends — forecasted job counts, remote-work growth, industry growth by company size
- AI Career Insights / Recommendations — personalized role, salary, and learning-path recommendations

2.3 Headline Metrics (as displayed)

Metric	Value shown
Total Jobs (Overview page)	151.445K
Total Companies	97
Employment Types	4
Total Jobs (Job Market Overview page)	78,726
Total Jobs (Future Trends page)	422
Median Salary	\$147K
Average Salary	\$162.84K

Data Coverage	work_year 2020 – 2025
Last Updated	15 May 2025

3. Page-by-Page Insights

3.1 Overview Page

- Clear value proposition stated up front — four supporting tiles (hiring trends, career roadmap, emerging tech, salary comparison, personalized recommendations) map directly to the five other pages, which is good UX practice.
- 97 companies across an estimated 151K job postings gives a reasonably broad market sample, though it's worth stating in the report/appendix which countries and job families are actually represented, since "AI job market" is a broad claim.

3.2 AI Job Market Overview

- Data Scientist (12.4%), Software Engineer (11.1%), Data Engineer (10%), and Data Analyst (9.1%) together account for roughly 43% of all roles — the market is still dominated by "core data" jobs rather than narrow AI specialisations (LLM engineer, MLOps, etc.), which is a genuinely useful insight for a student choosing where to start.
- Average salary climbs steadily from ~\$100K (2020) to a peak of ~\$160K (2024), then dips slightly in 2025 — worth calling out explicitly as "salary growth is flattening," since a plateau after four years of steep growth is itself a finding, not just a footnote.
- The insight callouts ("global demand continues to rise," "AI Engineer roles grew ~16%") are good — this is exactly the kind of one-line, decision-useful takeaway a dashboard should surface. Keep this pattern on every page.
- The "Average salary" and "Country" KPI cards on this page show implausible values (a 12-digit number, and a country count compared against a jobs goal) — see Section 6.1.

3.3 AI Career Roadmap

- The Beginner → Intermediate → Advanced skill ladder (Python/SQL/Excel/Statistics/Power BI → ML/Data Analysis/Deep Learning basics/Feature Engineering → Deep Learning/NLP/Computer Vision/MLOps/LLMs) is the strongest page in the dashboard. It's the one place where the project moves from "reporting data" to "giving advice," which is exactly what makes this different from a generic jobs-analytics dashboard.
- Job Hiring Trend shows near-flat postings through 2021, then a sharp inflection starting 2023 and accelerating into 2025 (0 → ~350) — this should be the headline chart of the whole project, since it's the clearest evidence for "why learn AI skills now."
- Most Demanded AI Job Roles reconfirms Data Scientist and Software Engineer at the top — pairing this directly next to the skill roadmap (which you've already done) is a strong design choice: it visually justifies why Python/SQL/ML sit at the base of the pyramid.

3.4 Salary Analysis

- Salary rises monotonically with experience level (Entry → Mid → Senior → Executive), which is expected but good to confirm — it validates that the dataset behaves the way a real compensation dataset should.

- Highest Salary is shown as \$30M — almost certainly a single outlier record (or a currency/unit conversion error) rather than a genuine AI salary. Left as-is, it undermines the credibility of every other number on the page. This is the single highest-priority fix in the whole project (see 6.2).
- The "Salary by Job Role" donut is labelled in billions ("3bn, 13.12%") — this is a sum of salary_in_usd across all records in that role, not a per-person salary, and the current label makes it easy to misread as an individual salary figure. Relabel as "Total Compensation Pool by Role" or switch the measure to average.
- The world map in this section plots values like 14,000–16,200 — far below the \$147K median shown at the top of the same page. That range looks like it belongs to a different field entirely (possibly a minimum-salary-by-country measure, or an unrelated column), and should be re-checked against the intended measure before this page is shown to anyone evaluating the model.

3.5 AI Future Trends

- Total Jobs on this page reads 422 — two orders of magnitude smaller than the 151K/78.7K shown on other pages. This is very likely a stale or unintended filter/slicer context carried over on this page rather than a true reflection of "future" jobs, and needs to be reconciled (see 6.1).
- "Fastest Growing Role: Web Developer" is an unexpected result to headline on an AI-focused dashboard — it may be accurate in the underlying data (broader tech-jobs dataset), but as the single largest, boldest number on the page it deserves a one-line explanation, or should be re-scoped to AI-specific roles only, to avoid confusing the reader about what the dashboard is actually about.
- Remote Work Growth (bar + line, 2020–2025) is one of the more convincing trend charts in the dashboard — a clear compounding rise from near-zero to ~1.5M+ by 2025 supports a genuinely useful takeaway: remote-friendly AI roles are becoming the norm, which matters for career planning.
- Industry Growth by company size shows Medium-sized companies as the dominant AI hirer, ahead of Large and (especially) Small companies — worth a one-line callout, since "mid-size companies are hiring AI talent fastest" is a non-obvious, useful finding for job seekers who default to assuming Big Tech is the main employer.

3.6 AI Career Insights / Recommendations

- Salary-by-title chart (Principal Data Architect down to Head of Marketing) is a strong long-tail view — it correctly shows a small number of leadership/architecture titles command dramatically higher average pay than the bulk of individual-contributor titles, which is a realistic and useful pattern to point out to someone planning a 5–10 year career arc.
- Experience-level donut (Senior 35.6%, Mid 31.3%, Entry 20.7%, Executive 12.4%) is a good complement to the salary-by-experience chart on the Salary Analysis page — consider merging these two views or at least cross-referencing them, since they tell the same story from two angles.
- The static "Employment_type: Part Time" and "Experience_level: Senior Level" cards look like they're meant to reflect a user's selected filter/persona, but with no visible slicer for role or persona on this page, a viewer can't tell whether these are global stats or a specific (possibly accidental) filter context — worth clarifying which it is.
- Learning Recommendations (Advanced Deep Learning, MLOps, System Design for AI, Cloud for AI/AWS) logically extend the Beginner/Intermediate/Advanced roadmap from page 3 — good continuity, though right now they appear to be static text rather than dynamically generated from the user's selected role/skills, despite the Overview page promising "personalized career suggestions based on your skills and preferences." Section 7 has a suggestion for closing that gap.

4. Cross-Page Key Findings

- **1. Core data roles remain the entry point into AI.** Data Scientist, Software Engineer, Data Engineer, and Data Analyst together make up over 40% of demand — the dashboard's own roadmap correctly starts learners here rather than jumping to LLMs/NLP.
- **2. Hiring has accelerated sharply since 2022–2023.** Both the salary trend and the job-hiring-trend charts show the same inflection point, which is strong triangulating evidence that 2023 marks a real structural shift in AI hiring, not noise in one chart.
- **3. Salary growth may be plateauing.** The 2024 peak followed by a 2025 dip (Job Market Overview page) deserves a dedicated one-line insight callout — right now it's visible in the chart but not called out in text anywhere, and it's a meaningfully different message from "salaries are rising."
- **4. Seniority still commands a very steep pay premium.** Executive vs Entry-level pay, and the long tail from Principal/Architect titles down to individual-contributor titles, both show the same 3–5x premium pattern for leadership/architecture tracks.
- **5. Remote work and mid-size companies are the two most "non-obvious" findings.** Both deserve more visual prominence than they currently get, since they're more actionable and less expected than "salaries are rising" or "Data Scientist is in demand."

5. What the Dashboard Does Well

- Consistent visual identity (color palette, robot mascot, left-nav pattern) across all six pages — this reads as a finished product, not a set of disconnected charts.
- A genuine narrative arc: Overview → Market → Roadmap → Salary → Trends → Personalized recommendation. Very few student dashboards go past "here is the data" into "here is what you should do about it."
- Good use of KPI cards with goal/variance formatting (green/red arrows) on the Job Market Overview page — this is a technique that shows familiarity with real BI reporting conventions, not just chart-building.
- The skill-roadmap page is a differentiator. Most public "AI jobs dataset" dashboards on Kaggle/GitHub stop at descriptive analytics; pairing that with a prescriptive roadmap is the strongest, most defensible part of this project in an interview.

6. Data Quality Issues to Resolve

These are the issues most likely to be raised by a technical reviewer or interviewer. None of them require new data — they are fixable with corrected DAX measures, filter context, and outlier handling.

6.1 Inconsistent "Total Jobs" across pages

The dashboard shows three different values for what appears to be the same underlying metric: 151.445K (Overview), 78,726 (Job Market Overview), and 422 (Future Trends). This is almost certainly caused by page-level filters/slicers, a disconnected date table, or a measure referencing a filtered table in one visual and the base table in another.

Recommendation: pick one canonical "Total Jobs" measure with an explicit, documented filter context, and reuse it everywhere; if pages are intentionally scoped differently (e.g., "Future Trends" = only 2025 postings), label the KPI card accordingly ("Total Jobs (2025 YTD)") so the difference reads as intentional, not broken.

6.2 Extreme salary outlier (\$30M "Highest Salary")

A single record (or a currency-conversion bug) is producing a \$30M salary figure. Recommendation: audit the top 1% of salary_in_usd, confirm currency conversion logic, and either correct or exclude clear data-entry errors; consider adding a capped/percentile view (e.g., 95th percentile) alongside the true max so extreme values don't distort the headline KPI.

6.3 "Average Salary" KPI showing a 12-digit number

On the Job Market Overview page, the Average Salary card displays a value in the billions (12,361,476,423), with a goal variance of +7,583,710.8%. This is a strong signal that the measure is summing salary_in_usd rather than averaging it, or that it's aggregating a percentage/ratio field incorrectly. Recommendation: re-check the DAX measure definition (AVERAGE vs SUM), and sanity-check every KPI card against its own visible unit before publishing.

6.4 Country KPI compared against a Jobs goal

The "Country" KPI card on the same page shows 8,668 against a goal of 78,726 (a -88.99% variance) — comparing a country count to a jobs target isn't a meaningful comparison and reads as a copy-pasted card whose goal wasn't updated. Recommendation: give this card its own goal (e.g., number of countries covered vs. total possible), or remove the goal/variance formatting entirely for count-of-distinct-values metrics.

6.5 Mismatched salary ranges on the map (Salary Analysis page)

The world map on the Salary Analysis page shows salary values of \$14,000–\$16,200, while the KPI cards on the same page show a \$147K median and \$162.84K average. These numbers look like they come from a different measure or a filtered subset (e.g., entry-level-only, or a different currency) rather than the same salary field. Recommendation: confirm the map's tooltip/value field matches the field used elsewhere on the page, and relabel if it is intentionally a different cut of the data.

6.6 Ambiguous "bn" (billions) labels on the Salary by Job Role donut

Because this visual is summing salary_in_usd across every employee in a role rather than averaging it, larger job categories will always look bigger regardless of actual pay level — this can mislead a reader into thinking Software Engineers are paid more than Data Scientists, when it may simply reflect there are more Software Engineer records. Recommendation: switch this specific visual to an average (or median) measure, since "per-person pay by role" is almost always the more meaningful comparison for a career-guidance tool than "total dollars paid to the role category."

7. Recommendations

7.1 Immediate Fixes (before presenting/submitting)

- Reconcile the Total Jobs figure across all three pages (or clearly label each as a different scope).
- Fix the Average Salary and Country KPI cards on the Job Market Overview page.
- Investigate and correct/exclude the \$30M salary outlier.
- Confirm the map values on the Salary Analysis page use the same field as the KPI cards on that page.
- Switch the Salary-by-Job-Role donut from sum to average/median, and relabel units clearly on every KPI card (K vs M vs bn).

7.2 Analytical Enhancements

- Add a dedicated "Insight" text box to every page (as already done on the Job Market Overview page) — one or two auto-generated or manually curated sentences per page keeps the dashboard readable for a non-technical stakeholder.
- Add drill-through from the roadmap page's "Most Demanded Job Roles" chart into a filtered Salary Analysis page for that specific role — this directly connects "what to learn" with "what it pays," which is currently something the reader has to do mentally across two pages.
- Add a simple trend/forecast line (even a straightforward linear or moving-average projection) to the Job Hiring Trend chart on the Roadmap page — this would make the "Future Trends" page feel materially different from a repeat of the same job-count time series shown on other pages.
- Consider a cohort or year-over-year % growth table instead of (or alongside) raw counts — a 16% YoY growth figure is more decision-useful than an absolute count, and you've already used this framing once in the Overview insight box.

7.3 Making "Personalized Recommendations" genuinely personalized

- Right now, Career Recommendation appears to show fixed text (fixed employment type, fixed experience level, fixed learning list) rather than reacting to user-selected filters. If time allows, add role/skill slicers that actually drive the Learning Recommendations list and the salary/role charts on that page — this would close the gap between the Overview page's promised feature ("personalized career suggestions based on your skills and preferences") and what the page currently delivers.

7.4 Future Roadmap for the Project (v2 ideas)

- Layer in a simple predictive model (e.g., regression or time-series forecast in Python/R, surfaced via a service or a pre-computed table) to project salary and hiring trends 1–2 years out, rather than only showing historical data through 2025.
- Add a skills-gap self-assessment: let a user check off which roadmap skills they already have, and surface only the missing ones as their personalized plan — a small but high-impact interactivity upgrade.
- Bring in a second data source (e.g., job-posting text) to extract in-demand tools/keywords (e.g., "LangChain," "PyTorch," "Snowflake") via simple text mining, to make the "Skills" page reflect live market language rather than a fixed curated list.
- Publish a short data dictionary / methodology note (source, date range, currency handling, definition of "AI job") as an appendix page or PDF — this pre-empts most of the data-quality questions a reviewer or interviewer is likely to ask (see the companion Q&A guide).

8. Conclusion

This is a well-conceived, well-designed project that goes beyond descriptive analytics into genuine career guidance — the roadmap and recommendation pages are the standout feature and the right thing to lead with when presenting it. The main work remaining is not new analysis but data hygiene: reconciling the Total Jobs figure, fixing two mis-scaled KPI cards, and resolving the salary outlier and map mismatch. Once those are addressed, this project is strong enough to anchor a portfolio, a resume bullet, or a technical interview conversation about end-to-end BI project ownership — including the ability to catch and explain your own data-quality issues, which is itself a skill interviewers look for.